



Feasibility of anaerobic digestion on the release of biogas and heavy metals from rice straw pretreated with sodium hydroxide

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Abstract

The feasibility of anaerobic digestion on the release of biogas and heavy metals from contaminated rice straw pretreated with NaOH solution was studied. The results show that NaOH pretreatment can significantly boost the release of biogas and heavy metals from rice straw using anaerobic digestion. Under the optimal conditions for biomass pretreated 6% (w/w) NaOH with a solid-to-solution ratio of 1:20, total biogas and methane yields of 446.3 mL/g and 263.5 mL/g volatile solids were achieved, which were 22.18% and 41.59% higher than those of the control without NaOH pretreatment, respectively, and the release percentages of Cd, Pb, Cu, and Zn from rice straw reached 86.95–97.69%. The release of heavy metals from rice straw can contribute to both the degradation of lignin by NaOH pretreatment and the utilization/transformation of lignocellulose via anaerobic digestion. The acidification levels and total volatile fatty acid contents significantly influence on the release of heavy metals. Based on the Illumina HiSeq sequencing analysis, the dominant phyla in the biogas residues were proteolytic (*Bacteroidetes*) and hydrogen-producing (*Firmicutes*) bacteria, while the growth of *Methanospirillum* and *Methanosaeta* in anaerobically digested effluent was promoted. The results revealed that anaerobic digestion combined with NaOH pretreatment is suitable for the disposal of heavy metal-contaminated biomass.

Keywords Rice straw · NaOH pretreatment · Anaerobic digestion · Heavy metals · Microbial community

Introduction

Over the last few decades, paddy soil in some regions has been contaminated potentially by toxic elements, such as cadmium (Cd), lead (Pb), and arsenic (As), which can be taken up by plants and pose risk to humans through the food chain (Han et al. 2018; Huang et al. 2018; Liu et al. 2018). However, high heavy metal contents in paddy soils increase the potential uptake of these metals by crops and produce heavy metal-contaminated rice straw (Feng et al. 2018). Therefore, the

disposal of heavy metal-contaminated rice straw has received great attention recently.

Open burning and incorporation are the most common traditional methods to dispose contaminated rice straw (Zhu et al. 2015). These unreasonable disposal technologies not only decrease the recovery of biomass resources but also seriously pollute the ecological environment (Negi et al. 2018). Anaerobic digestion is an attractive and plausible technology for rice straw disposal due to the significant advantages in sustainable energy generation, agricultural by-product recycling, and ecological sustainability (Chandra et al. 2012; Negi et al. 2018). Rice straw, composed of lignocellulose, is difficult for anaerobic microorganisms to degrade as it has a resistant structure composed of guaiacyl, syringyl, and hydroxyphenyl (Gao et al. 2017). Various pretreatment strategies, such as ultrasound, thermal, chemical, and biological methods, have been developed for the degradation of lignocellulose and increasing the accessibility of hemicelluloses to enzymes (Pellera et al. 2016; Gao et al. 2017). NaOH has been used to pretreat rice straw biomass, which can improve the removal of lignin and enhance the hydrolysis of lignocellulose (Mancini et al. 2017). A number of researches on anaerobic

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digestion, however, have exclusively paid attention to using rice straw for biogas production, but the release of toxic metal elements from straw biomass has rarely been reported. It is worth noting that the heavy metals are not biodegradable and long-term accumulative in anaerobic digesters, even if the contents of heavy metal in rice straw are considerably lower than the inhibition levels of stable digestion (Lee et al. 2017). Soluble Pb and Cd can adversely affect microbial activities (Nzihou and Stanmore 2013), owing to the interruption of enzyme structure and function by the formation of metal complexes with surface functional groups of protein molecules, such as thiol group, or by substituting naturally occurring metals in prosthetic enzyme groups (Valasatava et al. 2018). Many of the studies have only concerned about the inhibitory effect of soluble heavy metals on anaerobic digestion of rice straw, but fewer have focused on the release characteristics of heavy metals from rice straw during its digestion process.

Microbial community structure and diversity can affect the efficiency of rice straw digestion. The complete transformation of complicated organic matters to methane depends on syntrophic interactions between fermentative bacteria (hydrolysis, acidogenesis, and acetogenesis) and methanogens (Chen and Chang 2017). Under the Illumina sequencing platform, *Bacteroidetes*, *Proteobacteria*, and *Firmicutes* were revealed as the dominant phyla during the anaerobic digestion of dyeing wastewater (Xie et al. 2016). Similarly, *Firmicutes*, *Chloroflexi*, *Bacteroidetes*, and *Actinobacteria* were predominant under mesophilic conditions, while *Firmicutes*, *Thermotoga*, and *Synergistales* were far more abundant under thermophilic conditions during the anaerobic digestion of organic food waste (Lin et al. 2016). Thus, the effect of anaerobic digestion assisted by NaOH pretreatment on the changes in the microbial community's structure and diversity in biogas residue samples should receive great focus.

In this study, anaerobic digestion of contaminated rice straw pretreated with NaOH was carried out. The objectives of the study were aimed (1) to analyze the release of biogas and heavy metals from rice straw pretreated with NaOH solution via anaerobic digestion, (2) to investigate the interactions between digestion performance and the release of heavy metals from rice straw, and (3) to determine the effects of anaerobic digestion assisted by NaOH pretreatment on the community structures and diversities of fermentative bacterial and methanogenic archaeal.

Materials and methods

Raw straw and inoculum

Heavy metal-contaminated rice straw was obtained from a paddy field mainly contaminated by Cd in the Chang-Zhu-Tan region of Hunan Province. Rice straw was washed using

tap and deionized water successively. After oven-drying at 105 °C for 30 min and at 65 °C to constant weight, the straw was cut into small sections (20–30 mm) for pretreatment. A small amount of treated straw was completely smashed into pieces (0.250-mm sieve) so its components could be determined. The basic physicochemical properties of the raw straw are shown in Table 1.

The sewage sludge generated from a municipal wastewater treatment plant contains a large amount of the fermentative and methanogenic microorganisms. To enrich lots of microorganisms specialized in degradation of straw, the sludge should be acclimated according to the procedures reported by Xin et al. (2018). The sewage sludge contained 3.39 g P/kg sludge, 0.71 mg Cd/kg sludge, 7.63 mg Pb/kg sludge, 15.54 mg Cu/kg sludge, and 88.87 mg Zn/kg sludge. The pH value of the inoculum in the study, namely sewage sludge after acclimation, was adjusted to 7.50 ± 0.02 using 1 mol/L NaOH or HCl. Other physicochemical properties of the inoculum showed total solids of 17.24% and total carbon and nitrogen of 17.73% and 3.42%, respectively.

NaOH pretreatment

In this study, rice straw was pretreated with NaOH by soaking 50 g of rice straw in 1000 mL of 3%, 6%, and 9% (w/w) NaOH solution inside 2500-mL bottles, denoted as T1, T2, and T3, respectively (Table 1). The solid-to-solution ratio (g:mL) was 1:20. Rice straw soaked with deionized water without NaOH solution was used as the control treatment (CK). Each pretreatment was conducted in triplicate. The bottles were covered with plastic films, strictly sealed, and then incubated at 28 °C for 7 days. After pretreatment, the straw was filtered, cleaned, and then oven-dried at 65 °C to a constant weight. The pretreatment was then kept in a refrigerator so its composition could be determined and digested. The physicochemical characteristics of the pretreated rice straw are shown in Table 1.

Anaerobic digestion

A laboratory-scale simulated device was composed of 1-L anaerobic bottles, a thermostatic water bath, saturated brine displacement bottles, and graduated cylinders. The NaOH-pretreated straw was used as the digestion material, while the untreated straw was used as the control. The 20 g of pretreated or untreated rice straw and 7.0 g of the activated inoculum (wt.) were placed into anaerobic digestion bottles, followed by deionized water to obtain a total solid content of 8%. NaHCO_3 or $\text{CO}(\text{NH}_2)_2$ was utilized to adjust the C/N ratio to 20. The pH value of each bottle was adjusted to 7.50 ± 0.02 using 1 mol/L NaOH or HCl. All digesters were set in a thermostatic water bath at 37 °C for 36 days. The daily biogas yield was recorded once every 2 days through water displacement. The soluble

Table 1 Physicochemical characteristics of raw and pretreated rice straw

	Proximate and elemental analysis (%)				Content of heavy metals (mg/kg)				Content of components (%)		
	TS	VS	TC	TN	Cd	Pb	Cu	Zn	Cellulose	Hemicellulose	Lignin
Raw straw	94.17	87.80	55.29	1.63	2.20	2.90	7.88	77.88	35.28	35.31	16.87
CK (deionized water pretreated)	94.28	85.54	53.98	1.04	2.15	2.80	7.45	76.75	35.22	33.64	16.85
T1 (3% NaOH pretreated)	93.65	84.13	49.17	0.88	1.43	1.35	5.73	77.03	37.89	22.44	13.28
T2 (6% NaOH pretreated)	93.29	84.08	42.17	0.71	1.35	1.15	5.63	76.03	39.50	21.74	11.16
T3 (9% NaOH pretreated)	92.18	81.58	41.80	0.38	1.33	1.03	5.25	75.13	40.37	18.17	10.12

TS total solids, VS volatile solids, TC total carbon, TN total nitrogen

heavy metal contents attributed from digestion substrate (rice straw and inoculum) via digestion were investigated by dynamic sampling every 5 days. Each sample was digested in triplicate.

Sample analysis

Total solids (TS) and volatile solids (VS) were determined according to the methods described by Walter (2005). Total carbon (TC) and total nitrogen (TN) analysis of rice straw and inoculum (wt.) were performed by the elemental analyzer (vario EL III). The lignocellulose content measurements of rice straw were performed following the methods reported by Schultz et al. (1985). The biogas was collected by water displacement, and the gas volume was recorded once every 2 days. The CH₄ content was measured using a gas chromatograph mass spectrometer (GCMS-QP 2010 Ultra). The total volatile fatty acids (TVFAs) were analyzed following the colorimetric method, and the result was expressed in terms of the acetic acid content (Lee et al. 2012). The pH value, alkalinity (as CaCO₃), and soluble chemical oxygen demand (SCOD)/total chemical oxygen demand (TCOD) were measured according to APHA standard methods (Walter 2005). To determine the SCOD of anaerobically digested effluent, the soluble portion was isolated through a 0.45-μm filter (Walter 2005).

The ground straw samples (1 g, 0.250 mm) were digested by HNO₃ (10 mL) and HClO₄ (2 mL) following the instruction described by Liu et al. (2017). The inoculum and biogas residue samples were treated and digested according to the standard method of the soil samples (Zeng et al. 2017). The anaerobically digested effluent was further digested with HNO₃ following the detail operation procedure reported by Salem et al. (2014). The concentrations of Cd, Pb, Cu, and Zn in digested solutions were measured by ICP-OES (PerkinElmer Optima 5300DV). The lignocellulose structure was characterized using a FT-IR equipped with a DTGS detector (Nicolet IS10 instrument). The morphology and structure of the straw sample were observed through scanning electron microscopy (JSM-IT300LA) at a 10.0 kV acceleration voltage.

Statistical analysis

The release percentages (*R*) of Cd, Pb, Cu, and Zn from the rice straw via anaerobic digestion are calculated as follows:

$$R = (C_i \times M_i - C_r \times M_r) / C_i \times M_i$$

where *C_i* and *M_i* are the specific heavy metal content (mg/kg) and dry weight (mg) of the original rice straw before anaerobic digestion, respectively, while *C_r* and *M_r* are the specific heavy metal content (mg/kg) and dry weight (mg) of rice straw after anaerobic digestion, respectively.

Excel 2013 and Origin 8.5 were used for the data analysis and chart making. Redundancy analysis (RDA) was carried out to determine the multivariate relationship between the soluble metal contents and digestion performance indices during the 36 days of anaerobic digestion using CANOCO (version 4.5). One-way analysis of variance (ANOVA) was conducted to examine statistically significant differences using SPSS 13.0. Significant differences between means were determined by Duncan's test. The 5% level was considered to be statistically significant.

Microbial community analysis

DNA extraction

The acclimated inoculum and biogas residue after the 36-day digestion period (CK, T1, T2, and T3) were centrifuged at 10,000g for 10 min. The total microbial DNA was extracted from each sample using the Power Soil DNA Isolation Kit (Mo Bio Laboratories) in triplicate following the manufacturer's instruction. The extracted DNA was stored at −80 °C for further analysis.

PCR amplification and Illumina HiSeq sequencing

The hypervariable V3–V4 region of the bacterial 16S ribosomal RNA (rRNA) gene was amplified by a set of primers, 338F (5'-ACTCCTACGGGAGGCAGCAG-3')/806R (5'-GGACTACHVGGGTWTCTAAT-3') (Yu et al. 2005). The

primer pairs of Arch524F (5'-TGYCAGCCGCCGCGGTAA-3')/Arch958R (5'-YCCGCGCTTGAVTCCAATT-3') were used to amplify the 16S rRNA genes of methanogenic archaea (Pires et al. 2012). The detailed running procedures for PCR amplification were conducted as described by Xiao et al. (2017).

High-throughput sequencing analysis of rRNA genes was conducted on the purified, pooled sample using the Illumina HiSeq 2500 platform (2 × 250 paired ends). Operational taxonomic unit (OTU) numbers were recalculated with a 97% similarity cutoff (Li et al. 2016). The observed OTU, Chao1, the abundance-based coverage estimator (ACE), as well as the Simpson and Shannon diversity indices were estimated as the indicators to calculate microbial richness and diversity. Each sample analysis was conducted in triplicate.

Results and discussion

Effect of NaOH pretreatment on biogas production from rice straw

Cumulative biogas production has been recognized as a crucial indicator for determining the activity of methanogenic microorganisms (Xin et al. 2018), which can indicate the reaction stage of the digestion process. As shown in Fig. 1, the contents of NaOH for the pretreatment of rice straw notably influenced the cumulative and total biogas production but slightly affected the digestive process and variations in the daily biogas production. This was inconsistent as NaOH pretreatment could decrease the retention time of anaerobic digestion (Lin et al. 2009). The total biogas and methane yields of the NaOH-pretreated rice straw (T1, T2, and T3) were much higher than those of untreated straw (CK). The total biogas and methane yields of 446.3 mL/g and 263.5 mL/g VS in T2 with 6% NaOH were 22.18% and 41.59% higher than those of CK, respectively. The pretreated straw used as a digestion substrate in the treatment of T2 had a lignin content of 11.16%, which was 33.8% lower than that of untreated straw. A previous study demonstrated that lignin is degraded by a NaOH solution, and hemicellulose or cellulose is easily released and combined through enzymatic action, which led to enhanced hydrolysis and higher biogas production (Zhou et al. 2012). From comparing the treatment of T3 to that of T2, excessive alkalinity may cause a great loss in hemicellulose, which is a primary source of carbon for anaerobic digestion, thereby leading to a reduction in the total biogas and methane production (Song et al. 2014a, 2014b).

Changes of heavy metal contents in anaerobically digested effluent

In the early stage of anaerobic digestion, the contents of Cd and Pb were barely observed and below the detection level.

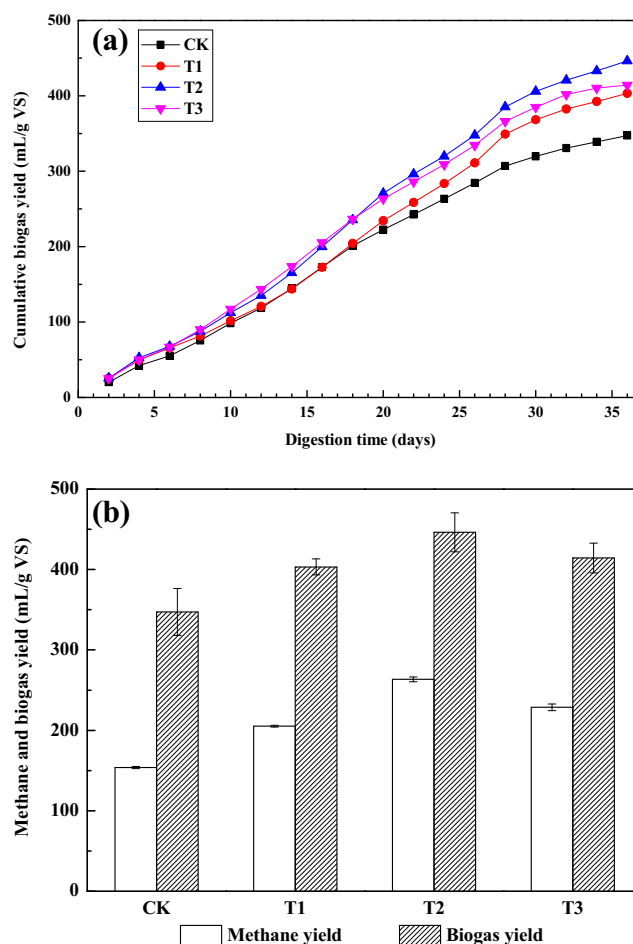


Fig. 1 Biogas and methane production from rice straw pretreated with NaOH via anaerobic digestion. **a** Cumulative biogas yield. **b** Total biogas and methane yield

During the whole experimental periods, little Cd and Pb were observed, as they might have been removed from the liquid phase by absorption onto the digestion substrate or precipitation with sulfur and hydroxide (Chen et al. 2008). The contents of Cu and Zn in the anaerobically digested effluent, however, changed significantly (Fig. 2). During the first 12 days of digestion, soluble metal contents changed unsteadily owing to the release of heavy metals from the inoculum. During days 12 to 31, the soluble contents of Zn and Cu remained relatively stable. From days 31 to 36, the contents of Zn and Cu increased significantly. The final contents of soluble metals in all treatments were 0.04–0.70 mg/L for Cd, 0.03–0.10 mg/L for Pb, 0.05–0.15 mg/L for Cu, and 0.15–0.37 mg/L for Zn, respectively, which were much lower than the reported inhibition levels and, therefore, might enter the enzymatic tissues which could drive numerous and essential digestion reactions (Chen et al. 2008; Yenigã et al. 2010). If the heavy metal contents of the anaerobically digested effluent exceed the inhibitory level, they could disrupt the enzyme function and structure by combining with thiol and other groups on protein molecules or replacing naturally occurring

metals in prosthetic enzymatic tissues (Lee et al. 2016). In brief, the contents of heavy metals had very little effect on anaerobic bacterial activities and the digestion process for biogas production.

High percentages of released heavy metals in rice straw pretreated with NaOH were achieved via anaerobic digestion (Fig. 3), which reached 81.49–96.08% for Cd, 55.11–89.55% for Pb, 55.77–90.84% for Cu, and 65.01–92.82% for Zn. The results indicated that significant amounts of heavy metals could be released from rice straw into anaerobically digested effluent via anaerobic digestion. As reported in previous work, Cu^{2+} could enhance the hydrolysis of lignin and hemicellulose, the degradation of non-volatile fatty acid (VFA) organic molecules, and the VFA composition and promote biogas production (Hao et al. 2017). The release percentages in treatments T2 and T3 were larger than those in CK and T1. This indirectly demonstrates that the pretreatment of rice straw with NaOH affected the release of heavy metals by increasing the degradation of lignocellulose and biogas production via anaerobic digestion.

Changes in digestion performance during anaerobic digestion of rice straw

TVFAs

VFAs are the main intermediate products and key indicators of the stability of anaerobic digestion (Alejo et al. 2018). During the 36-day digestion period, the TVFA content of the anaerobically digested effluent remained below the inhibition level (Fig. 4a) (under 6000 mg/L) reported by other researchers (Siegert and Banks 2005). The TVFA contents of all treatments were significantly higher during the first 5 days of digestion, when hydrolysis/acidification was most rapid. Peak values of 1584 mg/L in T2, 1394 mg/L in T3, 1344 mg/L in T1, and 1016 mg/L in CK were achieved in

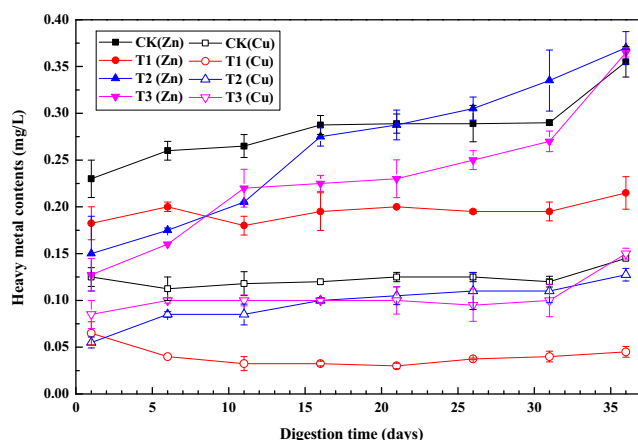


Fig. 2 Changes in the Cu and Zn contents of anaerobically digested effluent via anaerobic digestion

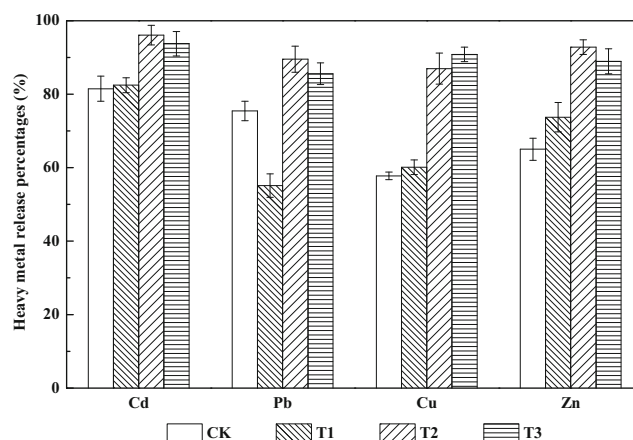


Fig. 3 Heavy metal release percentages from rice straw (wt.) pretreated with NaOH via anaerobic digestion

day 6. The TVFAs generated on the following days were utilized by methanogens; thus, their levels decreased dramatically (Shi et al. 2017). The TVFA contents in treatments T2 and T3 were higher than those in T1 and CK, indicating that more organic acids from the digesters in T2 and T3 were produced and utilized to generate biogas. The results are in accordance with another study where by monitoring the relevant increase of the TVFA concentration, the alkaline pretreatment enhanced the hydrolysis of the rice straw, which consequently caused a higher biogas production from the pretreated straw (Mancini et al. 2017).

Alkalinity and pH

Alkalinity and pH are closely connected with the TVFA contents in digesters (Cao et al. 2015). It is thought that the accumulation of TVFAs during anaerobic digestion is the primary reason of digester instability (Kamali et al. 2016), which could lead to the decline of pH value and alkalinity loss (Fig. 4b). The pH of all treatments remained relatively stable at 7.30–7.99 and did not significantly differ. The pH range of 7.37–7.79 in treatment T2 was more conducive for the degradation of rice straw and biogas production during anaerobic digestion (Anderson and Yang 1992). Alkalinity could be considered as a more reliable parameter to assess the stability of digesters than pH, because it decreased significantly before the pH decreased when TVFAs accumulated (Guwy et al. 1997). During the 36-day digestion period, alkalinity was not significantly correlated with NaOH pretreatment, as shown in Fig. 4c. The slight increase in alkalinity may be due to the activity of methanogenic microorganisms, which could produce alkalinity, such as carbon dioxide, ammonia, and bicarbonate (Lin et al. 2009). The above results suggest that the digestion performance of rice straw remained relatively stable and was not inhibited by extraneous toxic elements, especially heavy metals.

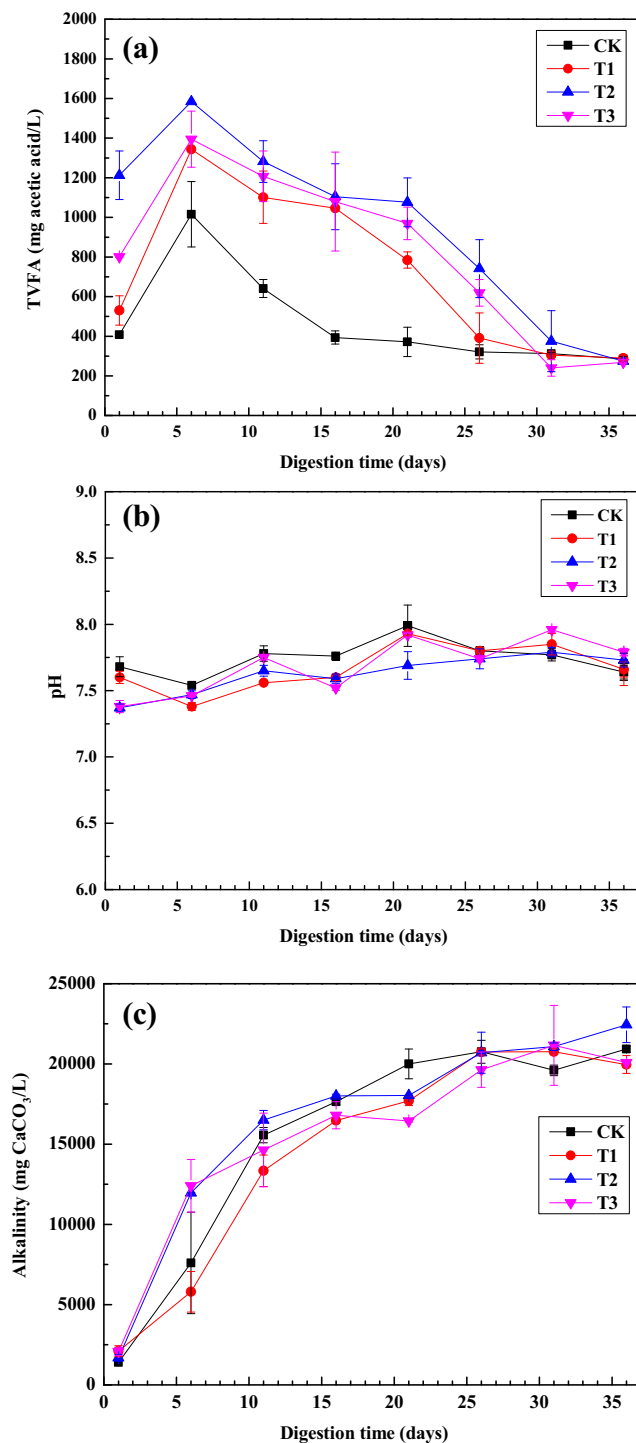


Fig. 4 a–c Stability of the anaerobic digestion of rice straw pretreated with NaOH

Hydrolysis and acidification

Hydrolysis and acidification are two pertinent indicators for evaluating the performance of digestion. The degree of hydrolysis is the ratio of SCOD to TCOD (SCOD/TCOD) (Demirel and Yenigun 2010). The degree of acidification is the ratio of

TVFA to SCOD (TVFA/SCOD) (Rubia et al. 2009). Hydrolysis and acidification of all treatments slightly differed during the 36-day digestion period (Supplementary data, Fig S1). The hydrolysis curves stably ranged from 0.61 to 0.91 at all treatments. From days 6 to 21, the degree of hydrolysis in CK was significantly lower than that of the others (T1, T2, and T3). The hydrolysis curve of T2 was almost always higher than that of the others, indicating that the hydrolysis of rice straw was influenced by the content of NaOH. The biogas and methane production of T2 was higher than those of the other treatments. Therefore, high levels of hydrolysis were observed with micromolecular soluble organic matters, such as oligomers or monomers, fatty acids, lactate, and alcohols, which were converted into methane via acetogenesis and methanogenesis (Orozco et al. 2013; Da Silva et al. 2015).

The ratio curve of TVFA/SCOD steadily declined during the 36-day digestion period (Supplementary data, Fig S1). Hydrolysis and acidification provided substrates for methanogens, which was also the main reason for the decrease in TVFA/SCOD during digestion. The acidification curve in T2 was lower than that of the others, while its TVFA curve was relatively higher (Fig. 4a). This indicates that the generated TVFAs had been fully utilized for methanogenesis. Hydrolysis was higher and acidification was lower in T2, which was pretreated with 6% NaOH, suggesting that complex organic macromolecules (such as lignocellulose) were transformed into micromolecular soluble organic matters by extracellular enzymes from some facultative or obligate anaerobic bacteria for methanogenesis, enhancing the biogas production from rice straw via anaerobic digestion.

RDA

RDA is typically used to analyze the relationship between heavy metal contents and digestion performance indices under the 36-day anaerobic digestion period (Fig. 5). According to the RDA, the release of heavy metals from rice straw is positively correlated with biogas production (especially Cd and Zu, $p < 0.01$) and can be enhanced by the degradation and utilization of organic matters. According to the directions and lengths of arrows, the heavy metal contents of anaerobically digested effluent are closely and negatively related to performance indices such as SCOD/TCOD, TVFAs, and TVFA/SCOD. The acidification levels and TVFA contents significantly influence the release of heavy metals (especially Cd and Zu, $p < 0.01$). It is possible that high levels of acidification promote the separation of heavy metals from lignocellulose and enhance the bioavailability of heavy metals in anaerobically digested effluent (Bolan et al. 2003). The angles of the arrows indicate that there are significantly positive correlations among the release of Cd, Pb, Cu, and Zn, which may simultaneously release to anaerobically digested effluent.

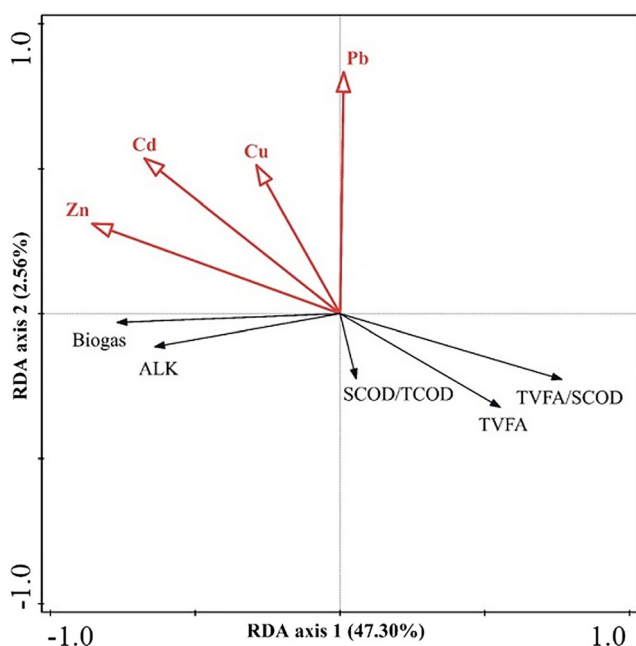


Fig. 5 RDA of soluble heavy metal contents with digestion performance indices via the anaerobic digestion of rice straw pretreated with NaOH (ALK alkalinity)

Degradation characterization of lignocellulose

The changes in the cellulose, hemicellulose, and lignin contents before and after pretreatment are shown in Table 1. The degradation percentages of hemicellulose and lignin in the pretreated rice straw were both increased with NaOH content. Especially, the degradation percentages of lignin for pretreated rice straw (T1, T2, and T3) were significantly higher than those of raw straw and CK ($p < 0.05$). After anaerobic digestion, the lignin content of rice straw was greatly degraded and reached 85.2% in treatment T2, while the total biogas and methane yields from the degraded lignocellulose were optimized in T2 and reached 446.3 mL/g and 263.5 mL/g VS, respectively. These results indicated that NaOH pretreatment significantly boosted the degradation of hemicellulose and lignin, while anaerobic digestion could enhance the utilization of lignocellulose.

The changes in the functional groups of untreated, 6% NaOH-pretreated, and digested straw (T2) were examined by FT-IR spectroscopy (Fig. 6). Comparing the raw straw with the pretreated and digested rice straw, lignin exhibited typical infrared spectra peaks of 1600 cm^{-1} (C=O, ester), $1430\text{--}1470\text{ cm}^{-1}$ (O-CH₃, methoxyl), and $700\text{--}900\text{ cm}^{-1}$ (C-H, aromatic hydrogen), indicating the absorbance of aromatic vibration, while the absorbance peaks at 1600 cm^{-1} and 1510 cm^{-1} varied, suggesting that NaOH pretreatment and anaerobic digestion reduced the lignin content.

The surface appearances of raw straw were observed using a scanning electron microscope. According to the SEM images of raw straw in Fig. 6a, the straw's structure contained

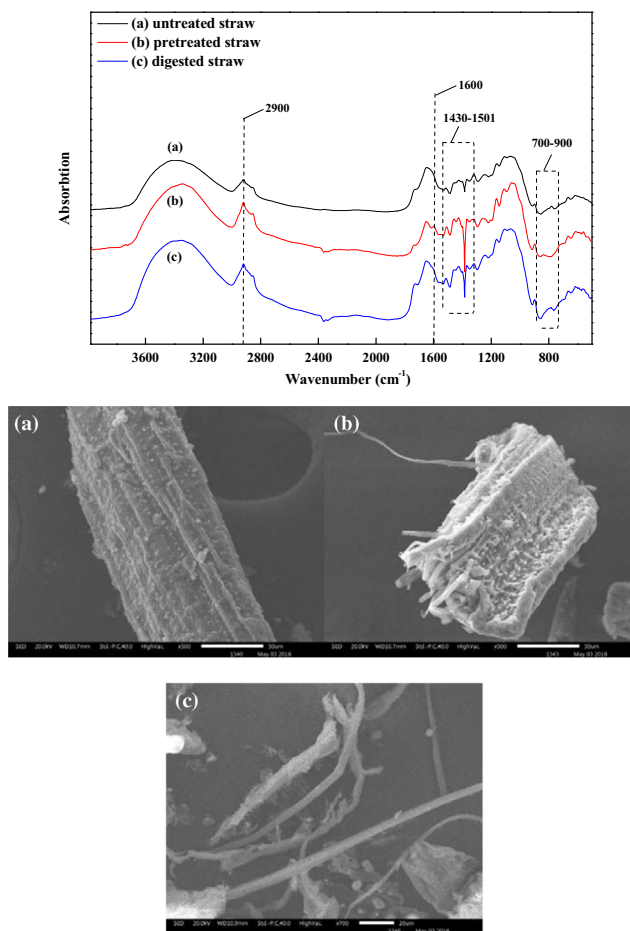


Fig. 6 FT-IR and SEM images of **a** untreated raw straw, **b** 6% NaOH pretreated rice straw before digestion, and **c** 6% NaOH pretreated rice straw after digestion

compact and stiff fibrils, which significantly hindered the degradation and hydrolysis of rice straw during anaerobic digestion (Guan et al. 2018). After NaOH pretreatment, the structures of lignin and hemicellulose were severely damaged and cellulose fiber was exposed (Fig. 6b), leading to an increase in the available surface area for contact with microbes and enzymes. Similar observations were found during alkaline pretreatment, which improved the accessibility of cellulose/hemicellulose with enzymes, thus enhancing methane production (Adl et al. 2012). The degradation percentages of hemicellulose and lignin at T2 relative to untreated raw straw were 41.1% and 85.2% via anaerobic digestion. The degradation of lignin in T2 was significantly higher than that in CK, T1, and T3.

Lignin has a high combination or absorption capacity with Cd, Pb, Cu, and Zn (Liang et al. 2013); thus, its degradation may enhance the release of heavy metals from rice straw. As shown in Table 1, NaOH pretreatment enhanced lignin degradation percentages, and the release of heavy metals increased with increasing NaOH content. Meanwhile, higher degradation percentages of lignocellulose, and release of biogas and

Table 2 Richness and diversity indices of bacterial communities in the inoculum and biogas residue samples via the anaerobic digestion of rice straw pretreated with NaOH

Sample ID	OTUs	ACE	Chao1	Simpson index	Shannon index	Good's coverage (%)
Inoculum	1282 ± 10	1340.8 ± 9.2	1386.5 ± 10.2	0.014 ± 0.003	5.59 ± 0.22	99.76 ± 0.04
CK	1001 ± 9	1248.1 ± 10.3	1307.8 ± 13.5	0.031 ± 0.007	4.51 ± 0.16	99.56 ± 0.05
T1	1033 ± 11	1262.2 ± 12.6	1280.1 ± 20.0	0.021 ± 0.003	4.77 ± 0.15	99.58 ± 0.04
T2	961 ± 9	1174.6 ± 15.9	1193.0 ± 15.0	0.062 ± 0.002	3.97 ± 0.10	99.62 ± 0.01
T3	989 ± 11	1178.4 ± 11.0	1198.4 ± 7.0	0.022 ± 0.001	4.78 ± 0.20	99.60 ± 0.02

heavy metals from rice straw were observed in T2 and T3, indicating that the degradation of lignocelluloses in rice straw, which were fully utilized and transformed into methane, caused heavy metals to secede and enter soluble forms in the anaerobically digested effluent.

Microbial community analysis

Bacterial community diversity and structure

After 36 days of digestion, the richness and diversity indices of the inoculum and biogas residue samples (CK, T1, T2, and T3) were analyzed by pyrosequencing to determine the changes in microbial communities via anaerobic digestion (Table 2). As indicated by the number of OTUs, Chao 1, and ACE, the richness of the bacterial communities was comparative in all samples, suggesting that NaOH pretreatment had a large influence on the species richness of bacteria. Good's coverage, which indicated sampling completeness, suggested that most bacterial phylotypes were detected in this study. The observed numbers of OTUs were 1282 (inoculum), 1001 (CK), 1033 (T1), 961 (T2), and 989 (T3), indicating that the richness of the bacterial community of biogas residue samples was lower than that of the inoculum. This demonstrated that partially dominant species were enriched after long-term digestion, while some bacteria were eliminated during anaerobic digestion operation.

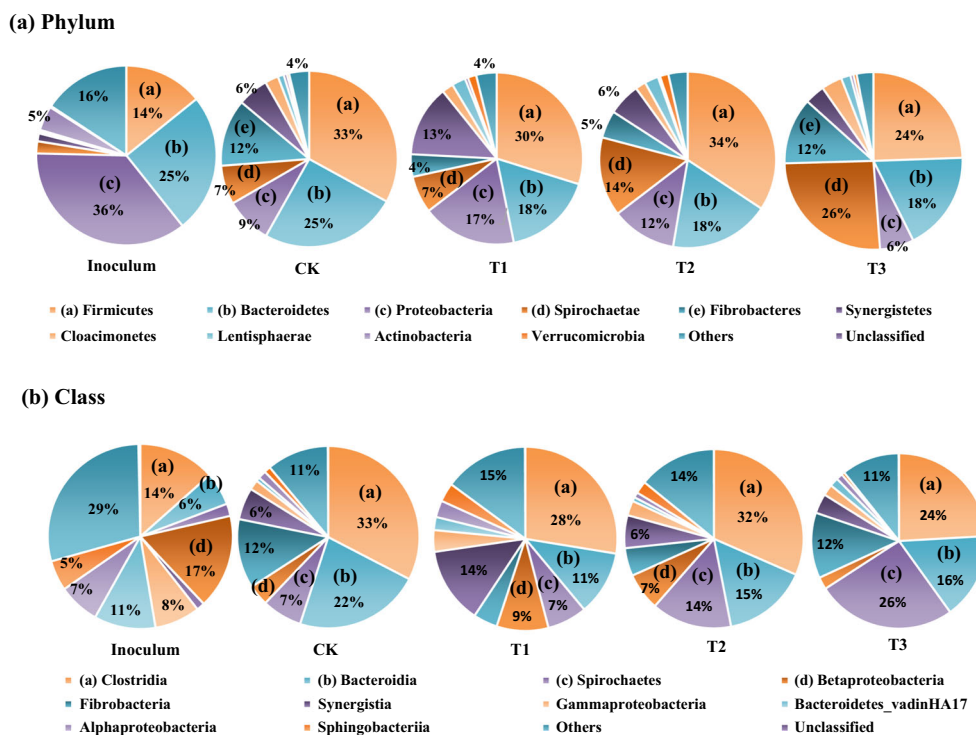
The diversity was estimated using the Shannon and Simpson indices on the OTU data. The bacterial diversity was positively correlated with the Shannon index, indicating the species abundance and negatively correlated with the Simpson index (Song et al. 2014a, 2014b). The Shannon index ranged from 3.97 to 5.59, with a higher value in the inoculum than the other samples (Table 2). The Simpson index varied among 0.014–0.062, with a higher value in T2 than the other samples and also indicated a decreasing trend with the increasing release of both biogas and heavy metals.

The phylogenetic classification of bacterial sequences in all samples at the phylum and class level is described in Fig. 7. The predominant phyla were *Firmicutes* (14.2–34.3%),

Bacteroidetes (17.0–25.2%), *Proteobacteria* (6.2–36.0%), *Spirochaetae* (2.2–25.8%), and *Fibrobacteres* (0.028–12.3%). As shown in Fig. 7a, *Firmicutes* were enriched, *Bacteroidetes* were slightly varying, and *Proteobacteria* were significantly eliminated with biogas production after anaerobic digestion. *Bacteroidetes* in all samples were in the largest proportion, which was frequently detected in anaerobic digesters and plays vital parts as fermenters and acidogens (Jang et al. 2014). As reported in previous studies (Yi et al. 2014), *Bacteroidetes* are proteolytic and may split many kinds of proteins into some products of small molecules, such as VFAs and NH₃. *Firmicutes* in samples CK, T1, T2, and T3 were enriched via anaerobic digestion and were 1.72–2.42 times richer than they were in the inoculum. *Firmicutes* are involved in the hydrolysis of polymers, such as cellulose and lignin, and produce organic acids as metabolic endpoints, which act as critical substances in acetoclastic methanogenesis (Yi et al. 2014). *Proteobacteria* constituted 36.0% of the community in the inoculum, while they only constituted 8.47%, 17.7%, 11.9%, and 6.16% of the communities in the biogas residue samples. *Proteobacteria* are common phyla to decompose complex organic matters, especially glucose and generated VFAs (Ariesyady et al. 2007).

The sequences classified at the class level were mainly composed of *Clostridia* (13.5–32.6%), *Bacteroidia* (5.8–22.4%), *Spirochaetes* (2.2–25.8%), *Betaproteobacteria* (2.3–16.6%), and *Fibrobacteria* (0.03–12.3%). Classes *Bacteroidia* and *Clostridia* belong to the *Bacteroidetes* and *Firmicutes* phyla, respectively, and were the dominant bacteria and more abundant in the samples (CK, T1, T2, and T3) than they were in the inoculum (Fig. 7b). Typically, *Bacteroidia* species can efficiently utilize complicated organic compounds and generated lactic or acetic acid into H₂ and CO₂ (Jang et al. 2014; Wirth et al. 2012). The *Clostridia* class contains several fermentative organisms which can decompose acetate butyrate and ethanol into hydrogen (Arreola-Vargas et al. 2013). The *Bacteroidia* and *Clostridia* classes were enriched by anaerobic digestion. The results suggested that these classes made a crucial difference in the solubilization of mesophilic sludge and VFA generation.

Fig. 7 Changes in the bacterial community structure at the **a** phylum and **b** class levels in the inoculum and biogas residue samples via the anaerobic digestion of rice straw pretreated with NaOH



Archaeal community diversity and structure

The composition of archaeal communities in the anaerobic digestion system was less complex than that of bacterial communities. The archaeal community in each sample contained fewer phylogenetically diverse groups, which was indicated by the existence of phylogenetically methanogenic groups, and accounted for a small proportion of the microflora in anaerobic digesters (Guo et al. 2015). The main phylum in all samples was *Euryarchaeota*, constituting 65.3–98.5% of the community. The predominant archaeal sequence at the class level was *Methanomicrobia* (19.8–49.4%), while *Methanomicrobiales* (20.0–78.9%) and *Methanosarcinales* (23.1–80.0%) were predominant at the order level. *Methanomicrobiales* were significantly more abundant in T1, T2, and T3 (77.0–80.0%) than they were in CK (20.0%), while *Methanosarcinales* were less abundant in T1, T2, and T3 than they were in CK. *Methanomicrobiales* was enriched, and *Methanosarcinales* lessened in T2, from which biogas production was optimal, than those of the other biogas residue samples. This indicated that methanogenic communities shifted from acetoclastic (*Methanosarcinales*) to hydrogenotrophic (*Methanomicrobiales*) methanogens with an increase in the proportion of syntrophic archaeal communities for biogas production via anaerobic digestion (Jang et al. 2014). *Methanospirillum* and *Methanosaeta* were prevailed, constituting approximately 96% of the total archaeal genera. *Methanospirillum*, which were hydrogen-utilizing methanogen in methane production, were more abundant in

T1, T2, and T3 than *Methanosaeta*. *Methanosaeta* can exclusively utilize acetate and H_2/CO_2 for methane production (Zhang et al. 2017). These results indicated that NaOH pretreatment could promote the growth of hydrogenotrophic (*Methanospirillum*) and acetate-utilizing (*Methanosaeta*) methanogens.

Conclusions

1. Pretreatment of rice straw with NaOH can facilitate biogas production and the release of heavy metals. A biogas yield of 446.3 mL/g VS was achieved, and 86.95–97.69% of heavy metals were released when rice straw was pretreated with 6% NaOH.
2. Anaerobic digestion combined with NaOH pretreatment significantly increased the biodegradation of lignocellulose, especially lignin, and greatly enhanced the release of heavy metals from rice straw. Acidification levels and TVFA contents significantly influenced the release of heavy metals. The contents of released heavy metals in the final anaerobically digested effluent were far below the inhibition levels and thus had little effect on stable anaerobic digestion.
3. Based on the Illumina sequencing analysis, the *Bacteroidetes* and *Firmicutes* bacteria and *Methanospirillum* and *Methanosaeta* archaea were predominant in biogas residue. The enriched fermentative bacteria

and methanogenic archaea fully utilized the organic matter to generate biogas and release heavy metals.

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